```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('/content/data of gurugram real Estate.csv')
df.sample(10)
```

	Price	Status	Area	Rate per sqft	Property Type	Locality	Builder Name	RERA Approval	BHK_Count	Socity	Company Name	Flat Type
12583	68900000.0	Ready to move	2420	28,512	3 BHK Apartment in Pioneer Presidia	Sector 62	HOMZ SOLUTIONS	Not approved by RERA	3.0	Pioneer Presidia	Pioneer	Apartment
7375	29800000.0	Under Construction	1401	21,300	3 BHK Apartment in Birla Navya Gurugram	Sector 63	WADHWA REALTORS	Approved by RERA	3.0	Birla Navya Gurugram	Birla	Apartment
1006	12500000.0	Ready to move	2000	6,250	3 BHK Independent Floor in Ansal Esencia	Sector 67	Sam Realtors	Not approved by RERA	3.0	Ansal Esencia	Ansal	Floor
					4 BHK							

```
df.shape
```

```
(19515, 12)
```

```
df.columns
```

```
Index(['Price', 'Status', 'Area', 'Rate per sqft', 'Property Type', 'Locality',
       'Builder Name', 'RERA Approval', 'BHK_Count', 'Socity', 'Company Name',
       'Flat Type'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19515 entries, 0 to 19514
Data columns (total 12 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   Price          19515 non-null  object 
 1   Status         19515 non-null  object 
 2   Area           19515 non-null  int64  
 3   Rate per sqft  19515 non-null  object 
 4   Property Type  19515 non-null  object 
 5   Locality       19515 non-null  object 
 6   Builder Name   19515 non-null  object 
 7   RERA Approval  19515 non-null  object 
 8   BHK_Count      19515 non-null  float64
 9   Socity         19515 non-null  object 
 10  Company Name   19515 non-null  object 
 11  Flat Type      19515 non-null  object 
dtypes: float64(1), int64(1), object(10)
memory usage: 1.8+ MB
```

```
df.describe()
```

	Area	BHK_Count
count	19515.000000	19515.000000
mean	2545.455752	3.229772
std	9550.629288	4.923873
min	60.000000	0.000000
25%	1423.000000	3.000000
50%	2000.000000	3.000000
75%	2800.000000	4.000000
max	958320.000000	132.000000

```
df.isnull().sum()
```

	0
Price	0
Status	0
Area	0
Rate per sqft	0
Property Type	0
Locality	0
Builder Name	0
RERA Approval	0
BHK_Count	0
Socity	0
Company Name	0
Flat Type	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(5292)
```

```
df = df.drop_duplicates()
```

```
df.isnull().sum()
```

	0
Price	0
Status	0
Area	0
Rate per sqft	0
Property Type	0
Locality	0
Builder Name	0
RERA Approval	0
BHK_Count	0
Socity	0
Company Name	0
Flat Type	0

dtype: int64

```
print("Status:")
print(df['Status'].value_counts())

print("\nProperty Type:")
print(df['Property Type'].value_counts())

print("\nRERA Approval:")
print(df['RERA Approval'].value_counts())

print("\nBHK:")
print(df['BHK_Count'].value_counts())

print("\nFlat Type:")
print(df['Flat Type'].value_counts())
```

```
Status:
Status
Ready to move         7823
Under Construction    5269
New                    834
Resale                 297
Name: count, dtype: int64

Property Type:
```

```
Property Type
3 BHK Independent Floor                                        392
3 BHK Apartment in M3M Golf Estate                             305
4 BHK Independent Floor                                        261
3 BHK Apartment in Krisumi Waterfall Residences                254
Residential Plot                                               242
                                                              ...
3 BHK Independent Floor in Bright Ace Palm Floors Apartment      1
2 BHK Apartment in Mapsko Paradise                               1
3 BHK Independent Floor in Sushant lok 2 Sector 57 Gurgaon       1
1 BHK Apartment in CHD Avenue 71                                 1
2 BHK Apartment in Reputed Builder Apna Enclave                  1
Name: count, Length: 1848, dtype: int64

RERA Approval:
RERA Approval
Not approved by RERA    8774
Approved by RERA        5449
Name: count, dtype: int64

BHK:
BHK_Count
3.0      6150
4.0      3723
2.0      2337
0.0       999
5.0       587
1.0       251
6.0        89
99.0       25
7.0        16
8.0        10
114.0       5
10.0        5
95.0        5
9.0         4
12.0        3
112.0       3
15.0        3
39.0        1
102.0       1
89.0        1
34.0        1
16.0        1
45.0        1
47.0        1
132.0       1
Name: count, dtype: int64
```

```
df[df['BHK_Count'] > 8][
    ['BHK_Count', 'Property Type', 'Flat Type', 'Area', 'Price']
].sort_values('BHK_Count')
```

	BHK_Count	Property Type	Flat Type	Area	Price
1825	9.0	9 BHK Independent House in Reputed Builder Sus...	House	5000	55000000.0
5103	9.0	9 BHK Independent House in Reputed Builder Sus...	House	2070	55000000.0
10811	9.0	9 BHK Apartment in Swaraj Homes Palam Vihar Ex...	Apartment	9500	25000000.0
18521	9.0	9 BHK Independent House	Plot	8100	8000000.0
15199	10.0	10 BHK Apartment in M3M Golf Estate	Apartment	15000	200000000.0
...	...	...	...	...	...
10759	114.0	Residential Plot in Emaar Business District Eb...	Plot	1163	38700000.0
10758	114.0	Residential Plot in Emaar Business District Eb...	Plot	1981	66000000.0
10757	114.0	Residential Plot in Emaar Business District Eb...	Plot	1130	37600000.0
10756	114.0	Residential Plot in Emaar Business District Eb...	Plot	969	32200000.000000004
19467	132.0	Residential Plot in I block 132	Plot	3240	57000000.0

61 rows × 5 columns

```
df[df['BHK_Count'] == 0][
    ['BHK_Count', 'Property Type', 'Flat Type', 'Area', 'Price']
]
```

	BHK_Count	Property Type	Flat Type	Area	Price
80	0.0	Residential Plot	Plot	1296	8500000.0
82	0.0	Residential Plot in Raheja Vanya City Plots	Plot	1602	22200000.000000004
87	0.0	Residential Plot in Faith Ireo Savannah	Plot	1485	13200000.0
94	0.0	Residential Plot in Meffier Golden Park	Plot	1575	13700000.000000002
140	0.0	Residential Plot	Plot	900	20000000.0
...	...	...	...	...	...
19466	0.0	Residential Plot in Vatika Express City Plots	Plot	1080	8500000.0
19468	0.0	Residential Plot in Vatika Express City Plots	Plot	1188	9500000.0
19470	0.0	Residential Plot	Plot	2016	14300000.0
19471	0.0	Residential Plot in Vatika Express City Plots	Plot	1080	8900000.0
19472	0.0	Residential Plot	Plot	900	620000.0

999 rows × 5 columns

```
df_clean = df.copy()

df_clean.loc[df_clean['BHK_Count'] > 10, 'BHK_Count'] = np.nan

df_clean['BHK_Count'].value_counts().sort_index()
```

BHK_Count	count
0.0	999
1.0	251
2.0	2337
3.0	6150
4.0	3723
5.0	587
6.0	89
7.0	16
8.0	10
9.0	4
10.0	5

dtype: int64

```
df_clean.isnull().sum()
```

	0
Price	0
Status	0
Area	0
Rate per sqft	0
Property Type	0
Locality	0
Builder Name	0
RERA Approval	0
BHK_Count	52
Socity	0
Company Name	0
Flat Type	0

dtype: int64

```
df_clean.duplicated().sum()
```

```
np.int64(0)
```

```
df_clean.dtypes
```

	0
Price	object
Status	object
Area	int64
Rate per sqft	object
Property Type	object
Locality	object
Builder Name	object
RERA Approval	object
BHK_Count	float64
Socity	object
Company Name	object
Flat Type	object

dtype: object

```
df_clean['Price'].sample(20)
```

	Price
1992	56000000.0
15804	57000000.0
16012	453200000.0
17387	35000000.0
16542	19800000.0
17806	80000000.0
15211	29300000.0
18533	36800000.0
7790	24000000.0
11422	35500000.0
14997	42100000.0
2445	23000000.0
8584	9279000.0
861	20500000.0
7004	75900000.0
11855	58300000.0
4554	19000000.0
14352	50000000.0
10015	293500000.0
13227	50500000.0

dtype: object

```
df_clean['Rate per sqft'].sample(20)
```

	Rate per sqft
16877	4,222
8044	10,666
2223	13,709
12497	3,997
14185	13,467
11764	13,714
7153	22,000
2785	11,949
13914	4,500
18699	5,647
14574	13,000
18215	5,500
11036	8,547
8632	2,20,000
1984	11,419
7491	17,016
340	13,000
351	10,810
17289	12,812
19465	7,870

dtype: object

```
df_clean['Price'] = pd.to_numeric(df_clean['Price'], errors='coerce')
df_clean['Rate per sqft'] = df_clean['Rate per sqft'].str.replace(',', '', regex=False)
df_clean['Rate per sqft'] = pd.to_numeric(  df_clean['Rate per sqft'], errors='coerce')
```

```
df_clean[['Price', 'Rate per sqft', 'Area', 'BHK_Count']].dtypes
```

	0
Price	float64
Rate per sqft	int64
Area	int64
BHK_Count	float64

dtype: object

```
df_clean[['Price', 'Rate per sqft', 'Area', 'BHK_Count']].isnull().sum()
```

	0
Price	1
Rate per sqft	0
Area	0
BHK_Count	52

dtype: int64

```
df_clean = df_clean.dropna(subset=['Price'])
df_clean[['Price', 'Rate per sqft', 'Area', 'BHK_Count']].isnull().sum()
```

	0
Price	0
Rate per sqft	0
Area	0
BHK_Count	52

dtype: int64

```
df_clean.loc[df_clean['BHK_Count'] > 10, 'BHK_Count'] = np.nan
df_clean['Price'] = pd.to_numeric(
    df_clean['Price'],
    errors='coerce'
)
# The 'Rate per sqft' column has already been converted to numeric in a previous cell.
# The .str accessor cannot be used on numeric types, which caused the AttributeError.
# Removing the redundant string replacement and keeping a single numeric conversion for robustness.
df_clean['Rate per sqft'] = pd.to_numeric( df_clean['Rate per sqft'], errors='coerce')
```

```
df_clean = df_clean.dropna(subset=['Price'])
df_clean[['Price', 'Area', 'Rate per sqft']].describe()
```

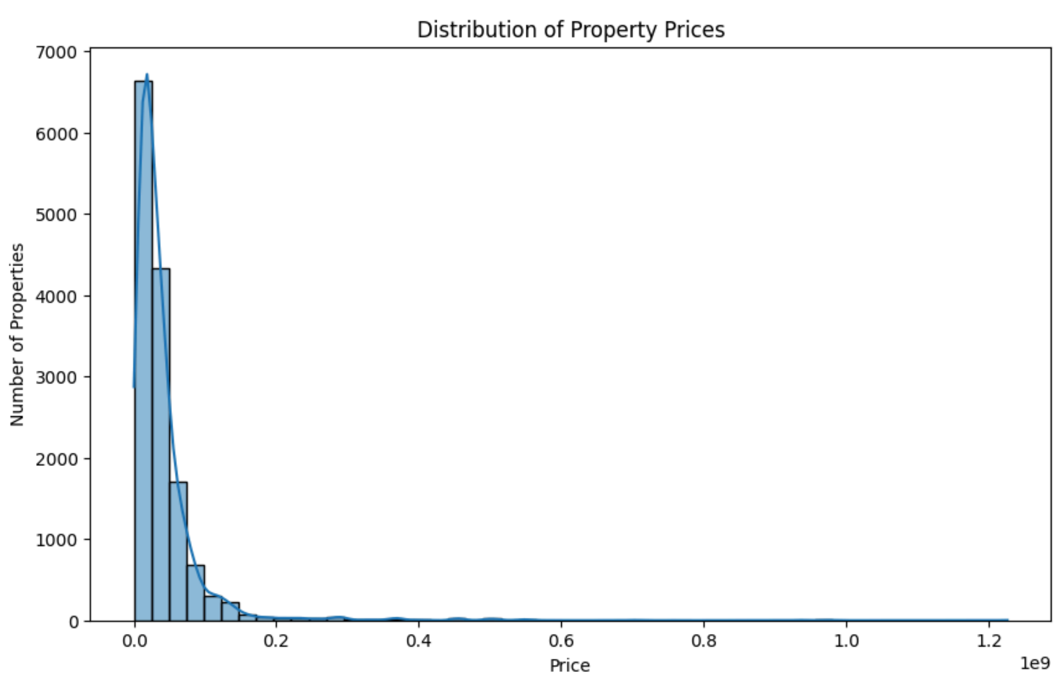
	Price	Area	Rate per sqft
count	1.422200e+04	14222.000000	14222.000000
mean	3.995657e+07	2621.927718	15297.182253
std	5.755396e+07	11125.670369	18761.253307
min	1.000000e+05	60.000000	100.000000
25%	1.410000e+07	1424.250000	8371.000000
50%	2.620000e+07	2015.000000	12380.000000
75%	4.550000e+07	2841.750000	18500.000000
max	1.226300e+09	958320.000000	310000.000000

```
plt.figure(figsize=(10, 6))

sns.histplot(df_clean['Price'], bins=50, kde=True)

plt.title('Distribution of Property Prices')
plt.xlabel('Price')
plt.ylabel('Number of Properties')

plt.show()
```

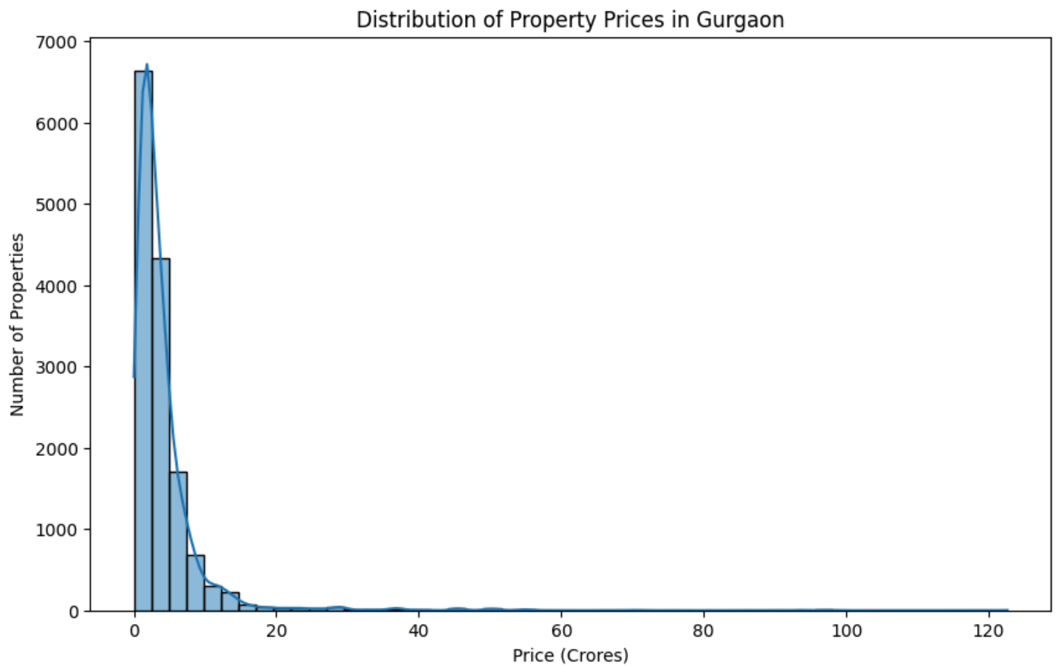


```
df_clean['Price_Cr'] = df_clean['Price'] / 10000000
df_clean[['Price', 'Price_Cr']].head()
plt.figure(figsize=(10, 6))

sns.histplot(df_clean['Price_Cr'], bins=50, kde=True)

plt.title('Distribution of Property Prices in Gurgaon')
plt.xlabel('Price (Crores)')
plt.ylabel('Number of Properties')

plt.show()
```



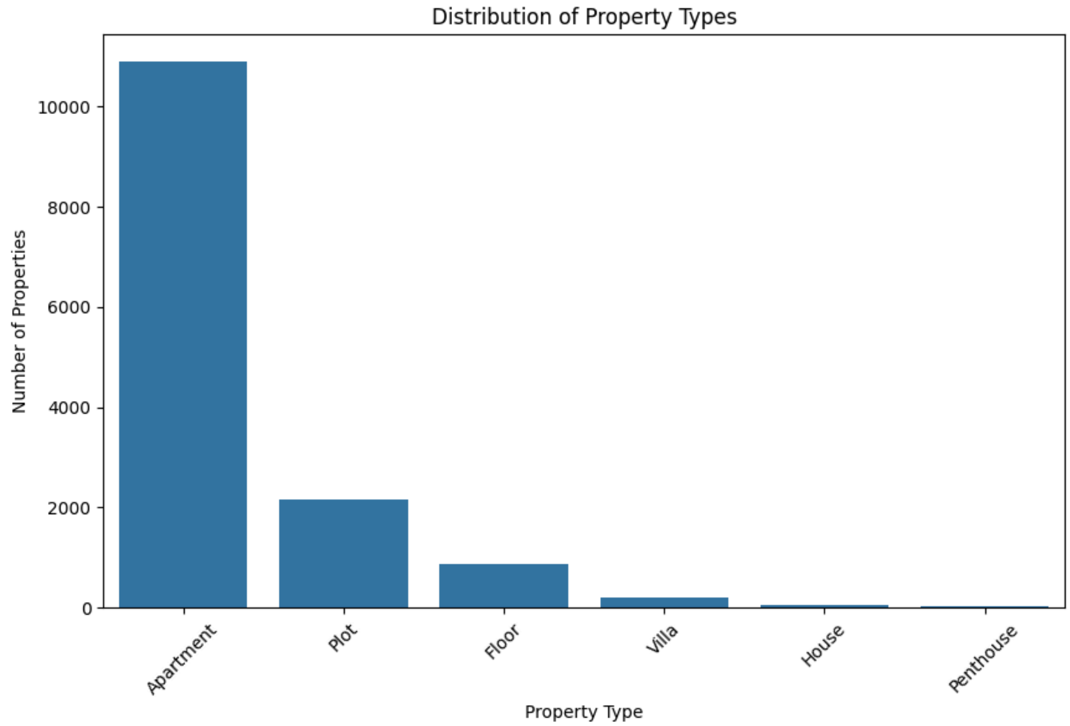

```
plt.figure(figsize=(10, 6))

sns.countplot(
    data=df_clean,
    x='Flat Type',
    order=df_clean['Flat Type'].value_counts().index
)

plt.title('Distribution of Property Types')
plt.xlabel('Property Type')
plt.ylabel('Number of Properties')

plt.xticks(rotation=45)

plt.show()
```



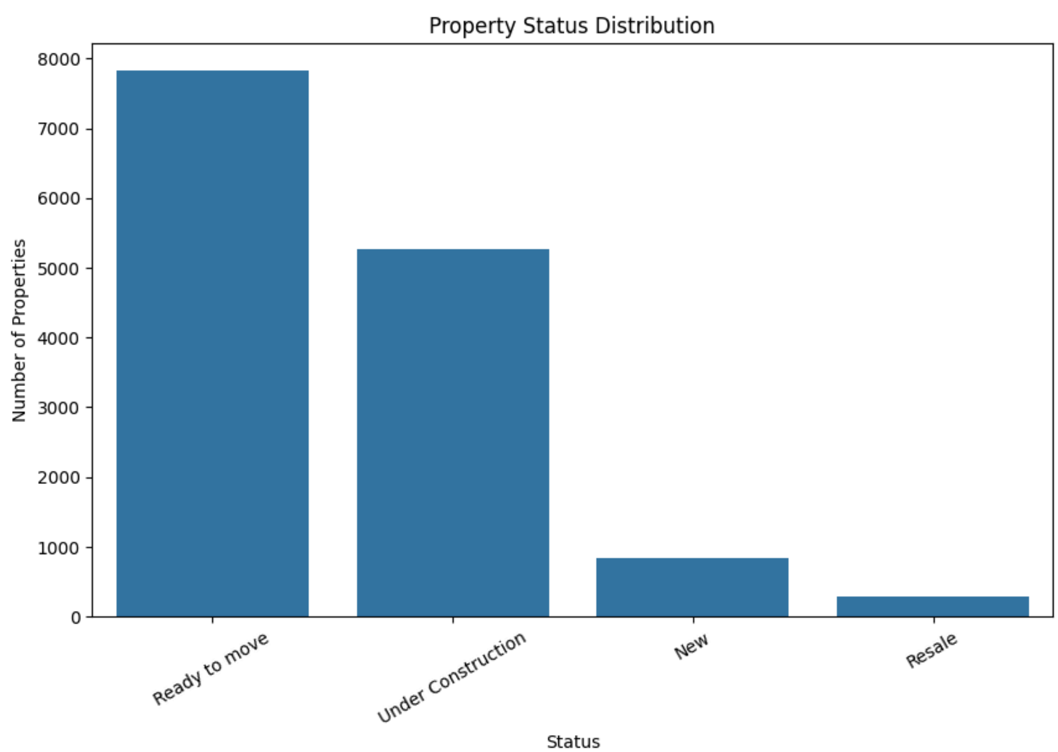

```
plt.figure(figsize=(10, 6))

sns.countplot(
    data=df_clean,
    x='Status',
    order=df_clean['Status'].value_counts().index
)

plt.title('Property Status Distribution')
plt.xlabel('Status')
plt.ylabel('Number of Properties')

plt.xticks(rotation=30)

plt.show()
```



```
bhk_price = df_clean.groupby('BHK_Count')['Price_Cr'].mean().reset_index()

bhk_price
```

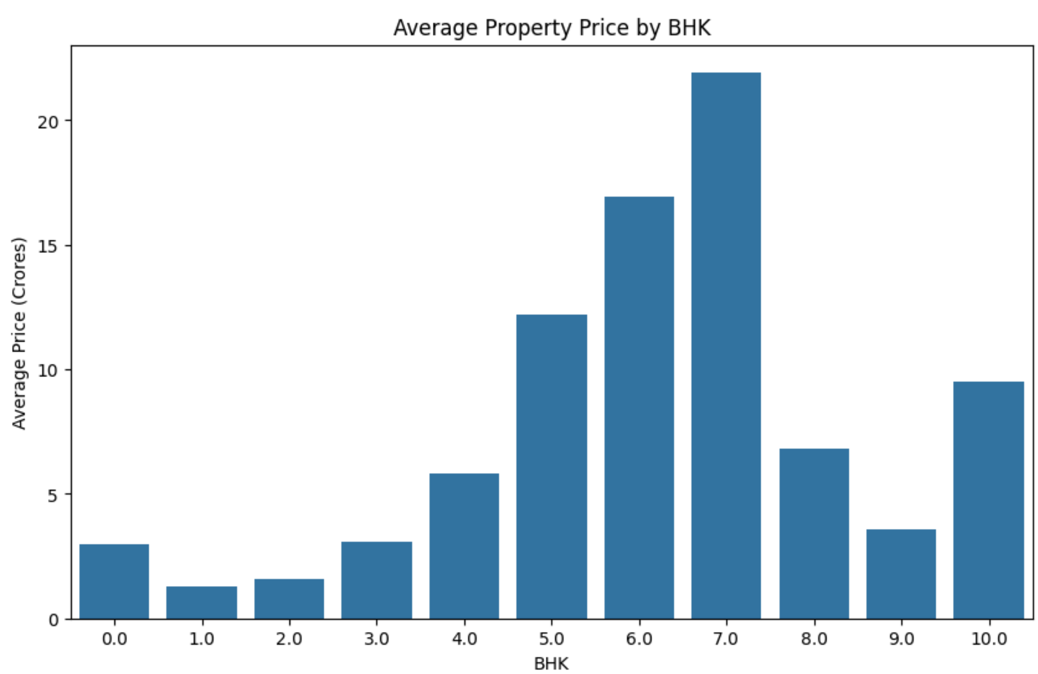
	BHK_Count	Price_Cr
0	0.0	2.974868
1	1.0	1.291090
2	2.0	1.565556
3	3.0	3.076041
4	4.0	5.818936
5	5.0	12.173612
6	6.0	16.913146
7	7.0	21.931250
8	8.0	6.801000
9	9.0	3.575000
10	10.0	9.500000

```
plt.figure(figsize=(10, 6))

sns.barplot(
    data=bhk_price,
    x='BHK_Count',
    y='Price_Cr'
)

plt.title('Average Property Price by BHK')
plt.xlabel('BHK')
plt.ylabel('Average Price (Crores)')

plt.show()
```




```
df_clean[df_clean['Area'] > 50000][
    ['Area', 'Price_Cr', 'Rate per sqft', 'Property Type', 'Locality', 'Flat Type']
].sort_values('Area', ascending=False)
```

	Area	Price_Cr	Rate per sqft	Property Type	Locality	Flat Type
17364	958320	11.97	125	Residential Plot	37D Dwarka Exp Gurgaon	Plot
17799	653400	75.00	1147	Residential Plot in Baliawas farms	Baliawas	Plot
17327	435600	50.00	1147	Residential Plot	Baliawas	Plot
17455	435600	50.00	1147	Residential Plot	Baliawas	Plot
12102	87300	9.00	1030	Residential Plot	Sector 75	Plot

```
normal_area = df_clean[df_clean['Area'] <= 10000]

plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=normal_area,
    x='Area',
    y='Price_Cr',
    alpha=0.5
)

plt.title('Area vs Property Price (Area ≤ 10,000 sq ft)')
plt.xlabel('Area (sq ft)')
plt.ylabel('Price (Crores)')

plt.show()
```

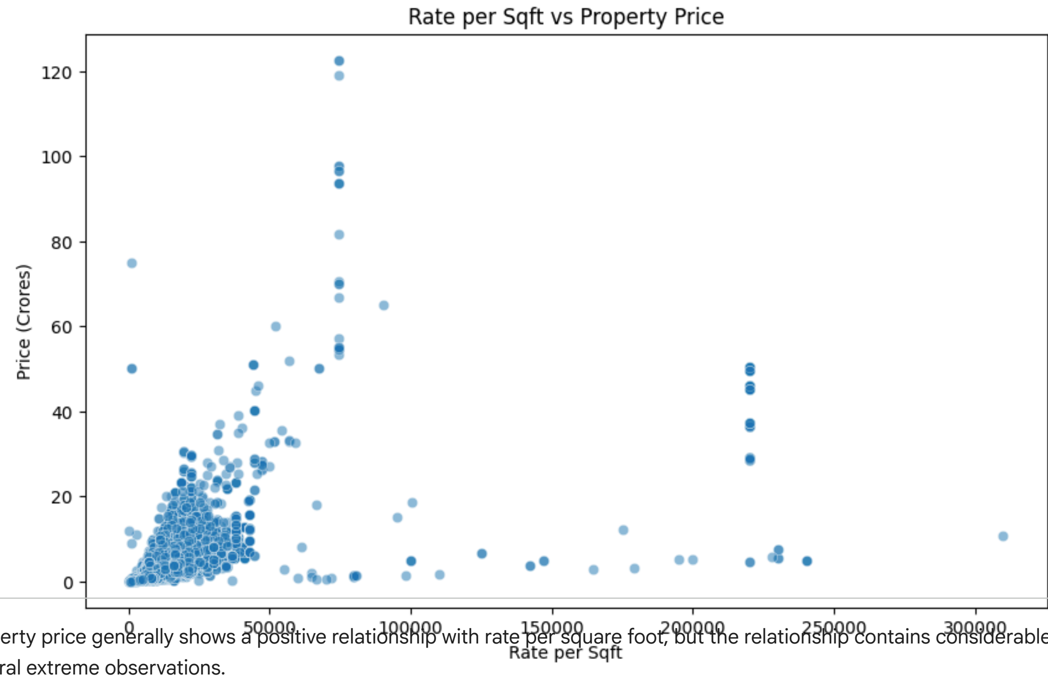
The analysis shows a positive relationship between property area and price.
Larger properties generally have higher prices, although significant variation exists, indicating that other factors such as location, property type and rate per square foot also influence property prices.

```
plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=df_clean,
    x='Rate per sqft',
    y='Price_Cr',
    alpha=0.5
)

plt.title('Rate per Sqft vs Property Price')
plt.xlabel('Rate per Sqft')
plt.ylabel('Price (Crores)')

plt.show()
```



Property price generally shows a positive relationship with rate per square foot, but the relationship contains considerable variation and several extreme observations.

```
numeric_cols = [
    'Price',
    'Area',
    'Rate per sqft',
    'BHK_Count'
]

correlation = df_clean[numeric_cols].corr()

plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation,
    annot=True,
    cmap='coolwarm',
```